How the numbers are calculated

Micro heat pipes and vapour chambers · calculations behind the slides · Debanshu Ghosh, 23CH3FP56. Each section gives the formula, one worked calculation you can repeat on a calculator, and the table the slide uses.

Slide	Numbers on the slide	Calculated in
3 (wick)	5× pump pressure, 25× liquid resistance	sections 2 and 3
4 (biporous)	50, 38, 161 W	section 5
5 (vapour core)	8× per halving; 1.08 K per kPa; 27×, 28×, 12×, ≈2×	sections 4 and 6

Everything on the slides comes from one script (calc.py, with its companion script for the extra steps), using standard steam-table properties of saturated water at 60 °C and the geometry in section 1. The chain is always the same: **inputs → formula → number → check**. Below, each step shows the formula, one worked calculation you can repeat on a calculator, and the table the slide uses.

1. Inputs

Quantity	Value	Kind of input
Surface tension σ	0.0662 N/m	standard steam-table value
Latent heat h_{fg}	2.358 MJ/kg	standard steam-table value
Liquid density, viscosity	983.2 kg/m³, 4.66×10⁻⁴ Pa s	standard steam-table values
Vapour density, viscosity	0.1304 kg/m³, 1.08×10⁻⁵ Pa s	standard steam-table values
Saturation pressure P_{sat}, T	19.95 kPa, 333.15 K (60 °C)	standard steam-table values
Heat load Q, source	5 W on 10 × 10 mm, so R1 = √(A/π) = 5.64 mm	assumed (phone-like, illustrative)
Chamber	disc with R2 = 48 mm, the same area as a 60 × 120 mm sheet	assumed
Wick layers, copper walls	2 × 50 µm wick, 2 × 80 µm wall	assumed
Porosity ε, r_{eff}	ε = 0.5; r_{eff} ≈ 0.21 d for sintered powder	assumed; rule of thumb
Cu mesh #200	r_{eff} = 64 µm, K = 2.8×10⁻¹¹ m²	assumed typical values
Contact angle	θ = 0° (full wetting)	assumed
Gravity head	Δz = 100 mm	assumed (vertical case)

Derived once and reused: ln(R2/R1) = ln(48/5.642) = 2.141, and the mass flow at 5 W, $\dot{m} = Q/h_{fg}$ = 5/2.358×10⁶ = 2.120e-06 kg/s.

2. The pump: capillary pressure (slide 3)

$$\Delta P_{cap} = \frac{2\sigma\cos\theta}{r_{eff}}$$

Worked: 50 µm powder, r_{eff} = 0.21 × 50 = 10.5 µm, so ΔP_{cap} = 2 × 0.0662 / 10.5×10⁻⁶ = 12.61 kPa. 10 µm powder: r_{eff} = 2.1 µm gives 63.0 kPa. Ratio = 5 (it is 50/10 = 5, because $\Delta P_{cap} \propto 1/d$).

3. The loss: permeability and liquid resistance (slide 3)

$$K = \frac{d^2 \varepsilon^3}{150(1-\varepsilon)^2}$$

$$\Delta P_l = a_l\, Q, \qquad a_l = \frac{\mu_l \ln(R_2/R_1)}{2\pi\, \rho_l\, K\, t_l\, h_{fg}}, \qquad t_l = 2t$$

Worked: 50 µm powder, K = (50×10⁻⁶)² × 0.125 / (150 × 0.25) = 8.333e-12 m² (that is d²/300). With t_l = 100 µm: a_l = 4.66×10⁻⁴ × 2.141 / (2π × 983.2 × 8.333e-12 × 1×10⁻⁴ × 2.358×10⁶) = 82.2 Pa/W. The same steps for 10 µm powder give 2055 Pa/W, a ratio of 25 (K ∝ d², so 25).

Wick	r_{eff} (µm)	K (m²)	ΔP_{cap} (kPa)	a_l (Pa/W)
Cu mesh (#200)	64.0	2.80e-11	2.1	24.5
Sintered, d = 50 um	10.5	8.33e-12	12.6	82.2
Sintered, d = 10 um	2.1	3.33e-13	63.0	2054.7

4. The vapour resistance and its temperature cost (slide 5)

$$\Delta P_v = a_v\, Q, \qquad a_v = \frac{6\,\mu_v \ln(R_2/R_1)}{\pi\, \rho_v\, h^3\, h_{fg}}, \qquad \Delta T_{sat} \approx \frac{T \Delta P}{\rho_v\, h_{fg}}$$

Worked: h = 100 µm, a_v = 6 × 1.08×10⁻⁵ × 2.141 / (π × 0.1304 × (10⁻⁴)³ × 2.358×10⁶) = 144 Pa/W. At 5 W: ΔP_v = 718 Pa. The conversion factor is T/(ρ_v h_{fg}) = 333.15 / (0.1304 × 2.358×10⁶) = 1.08×10⁻³ K/Pa, so ΔT = 0.78 K. Halving h to 50 µm multiplies a_v by 2³ = 8.

h (µm)	ΔP_v at 5 W (Pa)	ΔT at 5 W (K)	Mach	Knudsen	Valid? (ΔP < 20% P_{sat})
400	11	0.01	0.003	0.0019	yes
200	90	0.10	0.005	0.0038	yes
100	718	0.78	0.010	0.0077	yes
50	5745	6.22	0.020	0.0154	no: trend only

The 20% limit is ΔP_v = 0.2 × 19.95 kPa = 3.99 kPa; solving a_v Q = 3.99 kPa for h at 5 W gives h ≈ 56 µm. Below that the linearised relation is not trusted. Validity checks at h = 100 µm: Mach = u/√(γ R_v T) = 0.010 (u = 4.6 m/s, incompressible); Knudsen = λ/h = 0.77/100 = 0.0077 (continuum, no slip).

5. The capillary limit and the biporous bars (slide 4)

$$Q_{max} = \frac{\Delta P_{cap} - \rho_l g\, \Delta z}{a_l + a_v}$$

Worked, 50 µm powder with h = 100 µm (horizontal, Δz = 0): Q_{max} = 12610 / (82.2 + 144) = 56 W. The three biporous bars use the same equation with a different (r_{eff}, K) pair, all at h = 100 µm and 2 × 50 µm wick:

Case	r for ΔP_{cap}	K for a_l	ΔP_{cap}	a_l	a_v	Q_{max}
Fine only	5 µm	1.76e-12 (d = 23 µm)	26.5 kPa	388 Pa/W	144	50 W
Coarse only	21 µm	3.33e-11 (d = 100 µm)	6.3 kPa	20.5 Pa/W	144	38 W
Biporous	5 µm	3.33e-11	26.5 kPa	20.5 Pa/W	144	161 W

Check the biporous bar: 26480 / (20.5 + 144) = 161 W. The fine wick takes its permeability from a 23 µm powder because its pore radius is 0.21 × 23 ≈ 5 µm, so r and K are consistent with each other. The

biporous bar combines the fine r with the coarse K, which is why it is an idealised upper bound.

6. Chamber against a solid copper sheet (slide 5)

$$\Delta T_{Cu} = \frac{Q \ln(R_2/R_1)}{2\pi k H}, \qquad \Delta T_{chamber} = \Delta T_{vapour} + \Delta T_{wick} + \Delta T_{interface}$$

The copper sheet has the chamber's total thickness H = h + 2 × 50 + 2 × 80 µm and k = 400 W/m K. The chamber terms: vapour from section 4; wick conduction $\Delta T_w = q'' t_w / k_{wick}$ with $q'' = Q/A = 50000$ W/m² and $k_{wick} = 160$ W/m K from the Maxwell estimate for ε = 0.5; interface $\Delta T_i = q''/h_i$ with $h_i = 2.33 \times 10^5$ W/m² K from kinetic theory (Hertz–Knudsen–Schrage) with an assumed accommodation coefficient of 0.1.

Worked, h = 200 µm: H = 0.46 mm, $\Delta T_{Cu} = 5 \times 2.141 / (2\pi \times 400 \times 0.00046) = 9.3$ K. Chamber: 0.097 + 0.016 + 0.215 = 0.33 K. Ratio = 28.

h (µm)	H (mm)	Copper (K)	Vapour (K)	Wick (K)	Interface (K)	Chamber (K)	Copper ÷ chamber
400	0.66	6.5	0.012	0.016	0.215	0.24	27×
200	0.46	9.3	0.097	0.016	0.215	0.33	28×
100	0.36	11.8	0.778	0.016	0.215	1.01	12×
50	0.31	13.7	6.224	0.016	0.215	6.45*	≈2×*

** ΔP_v above 20% of P_{sat}, so a trend only. Why copper catches up: copper's ΔT rises only from 6.5 to 13.7 K as H falls from 0.66 to 0.31 mm, while the chamber's vapour term rises from 0.01 to 6.2 K because it goes as 1/h³.*